Côme Cattin

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Birth date : 24th April 2000 (25)

PROFILE

Theoretical chemistry PhD. student passionate about fundamental physical chemistry, machine learning, teaching, research and culture.

EDUCATION

2024 – Today **Sorbonne Université, PhD in Theoretical Chemistry**
Development of hybrid force fields combining neural networks and physical laws, applied to biological systems
Advisor : Jean-Philip Piquemal

2021 – 2024 **École Normale Supérieure de Lyon, Master Sciences of Matter, Computational Physics and Chemistry**

2020 **École Normale Supérieure de Lyon, Bachelor's degree Sciences of Matter**

2018 – 2020 **Université de Paris, double Bachelors's degree in Physics and Chemistry**

SKILLS

Chemistry

- *2024* - Internship at Qubit Pharmaceuticals and Sorbonne Université, Paris, France, Development of an hybrid machine learning/physical force field for molecular simulations.
- *2023* - 6 months internship at the Institut Lumière Matière, Lyon, France, Conformation dynamics study of the HSP90-NTD protein using molecular dynamics simulations and Markov State Models.
- *2022* - 3 months internship at the Max-Planck Institute For Solid State Research, Stuttgart, Germany, Exotic electronic phases in honeycomb materials. Solid state synthesis and characterization, quantum spin liquids.
- *2021* - 2 months intership at the Institute of Physical Chemistry (ICP), University of Orsay, France. Understanding the redox properties of cytochrome *bd* in various organisms using the Density Functional Theory. Use of quantum chemistry software (deMon2k, Gaussian).

Computer Science

- Coding in Python, Fortran, Matlab, Bash, HTML/CSS, JavaScript, Swift, SQL
- Use of machine learning libraries (Jax, PyTorch)
- LaTeX, Git, Docker, HPC

Teaching

- 2025 - Mathematical Tools for Physical Chemistry for second year chemistry students at Sorbonne Université
- 2025 - Career guidance and professional development courses for first year Physics and Chemistry students at Sorbonne Université
- 2024 - Practical general chemistry courses for first year chemistry students at Sorbonne Université

LANGUAGES

- Native French
- English C1 (Cambridge Advanced Exam)
- Italian B2

OTHER

Sports

- Climbing, swimming, boxing,

Music

- Guitar, piano, ukulele, mandolin

Publications

- Cattin, C., Plé, T., Adjoua, O., Gouraud, N., Lagardère, L., & Piquemal, J. P. (2025). Accelerating Molecular Dynamics Simulations with Foundation Neural Network Models using Multiple Time-Step and Distillation. arXiv preprint arXiv :2510.06562.
- Siddeeque, R., Etcheverry, B., Cattin, C., Deviers, J., Melin, F., Hellwig, P., ... & de la Lande, A. (2025). Computational study of heme b595 to heme d electron transfer in E. coli cytochrome bd-I oxidase. bioRxiv, 2025-09.